DSA06 – DATA HANDLING AND VISUALIZATION

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R Programming Lab – Assignment 1

Assignment No. 1 Course Outcome CO1

Title Fundamentals of Data Visualization: Aesthetics, Scales, Coordinates and Color

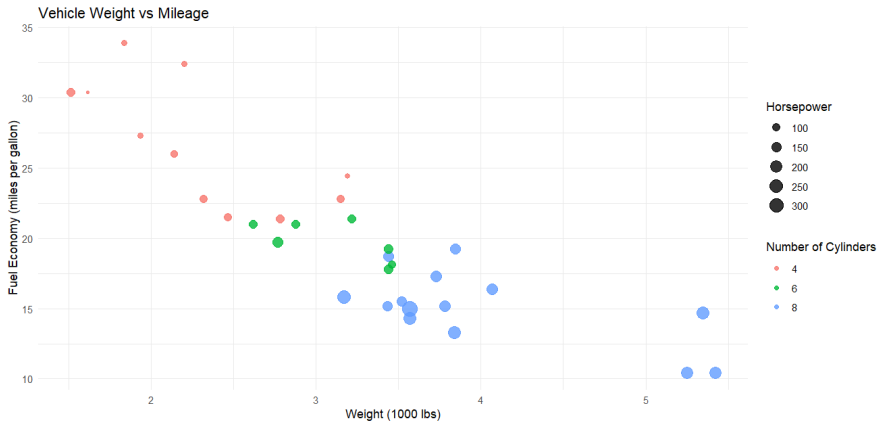
Q1. Mapping Data onto Aesthetics

(a) Scatterplot of wt vs mpg using mtcars

The scatterplot maps vehicle weight (wt) to the x-axis and mileage (mpg) to the y-axis. Number of cylinders (cyl) is represented by color, while horsepower (hp) is represented by point size. The title, axis labels, and legend titles are specified.

```
library(ggplot2)
ggplot(mtcars, aes(x = wt, y = mpg,
                   color = factor(cyl), size = hp)) +
  geom_point(alpha = 0.8) +
  labs(title = "Vehicle Weight vs Mileage",
       x = "Weight (1000 lbs)",
       y = "Fuel Economy (miles per gallon)",
       color = "Number of Cylinders",
       size = "Horsepower") +
  theme_minimal()
```

OUTPUT:



(b) Structure and classification of mpg variables

The mpg dataset from ggplot2 contains 11 variables. Quantitative variables measure numerical amounts; qualitative variables represent categories. Suggested aesthetics are shown below.

Variable	Type	Suitable aesthetic
manufacturer	Qualitative	color / shape
model	Qualitative	label / facet
displ	Quantitative	x / y / size
year	Quantitative (discrete)	x / y / color
cyl	Quantitative (discrete)	color / shape
trans	Qualitative	color / shape
drv	Qualitative	color / shape
cty	Quantitative	x / y / size

hwy	Quantitative	x / y / size
fl	Qualitative	color / shape
class	Qualitative	color / facet

```
library(ggplot2)
str(mpg)
```

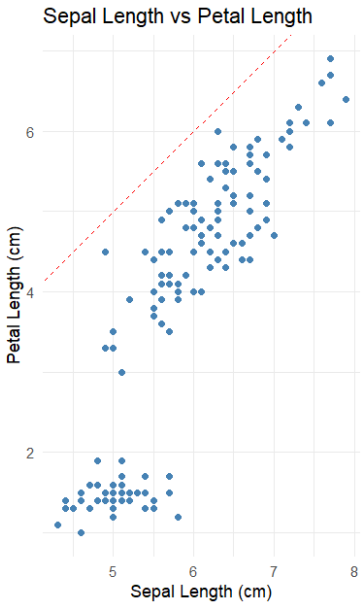
Q2. Coordinate Systems and Axes

(a) Iris scatterplot in Cartesian coordinates with aspect ratio 1

Sepal.Length and Petal.Length are measured in centimeters. coord_fixed(ratio = 1) gives equal physical scaling to the x- and y-units. The dashed y = x line is a reference.

```
library(ggplot2)
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length)) +
  geom_point(color = "steelblue") +
  geom_abline(slope = 1, intercept = 0,
              linetype = "dashed", color = "red") +
  coord_fixed(ratio = 1) +
  labs(title = "Sepal Length vs Petal Length",
       x = "Sepal Length (cm)",
       y = "Petal Length (cm)") +
  theme_minimal()
```

OUTPUT:

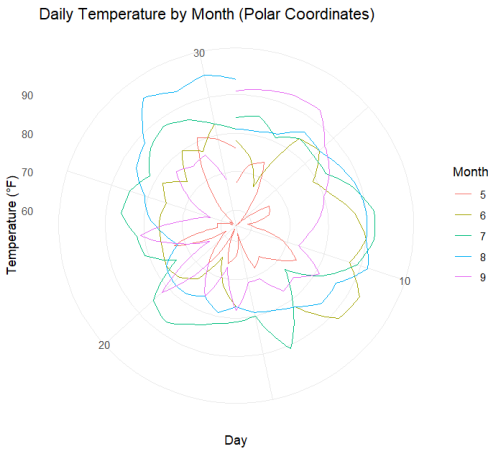


(b) Daily temperature by month in polar coordinates

The following plot displays daily temperature by day of month, with separate colored traces for each month. coord_polar() converts the day axis to an angular position.

```
library(ggplot2)
ggplot(airquality, aes(x = Day, y = Temp,
                       color = factor(Month), group = Month)) +
  geom_line() +
  coord_polar() +
  labs(title = "Daily Temperature by Month (Polar Coordinates)",
       x = "Day", y = "Temperature (°F)",
       color = "Month") +
  theme_minimal()
```

OUTPUT:

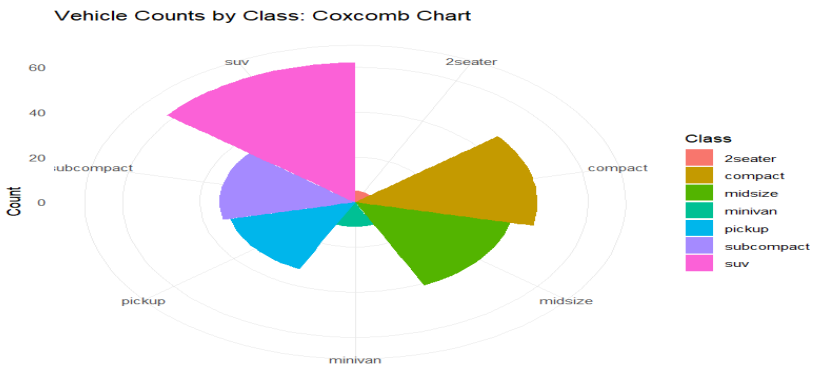


Polar coordinates are useful for cyclic or periodic data, such as time of day, day of year, seasonal patterns, wind direction, and repeated cycles. They can reveal wrap-around patterns, but are less suitable when precise linear comparisons are important.

(c) Bar chart and coxcomb chart of vehicle counts

```
library(ggplot2)
# Cartesian bar chart
ggplot(mpg, aes(x = class)) +
  geom_bar(fill = "skyblue") +
  labs(title = "Number of Vehicles by Class",
       x = "Vehicle Class", y = "Count") +
  theme_minimal()

# Coxcomb (polar bar) chart
ggplot(mpg, aes(x = class)) +
  geom_bar(aes(fill = class), width = 1) +
  coord_polar() +
  labs(title = "Vehicle Counts by Class: Coxcomb Chart",
       x = NULL, y = "Count", fill = "Class") +
  theme_minimal()
```



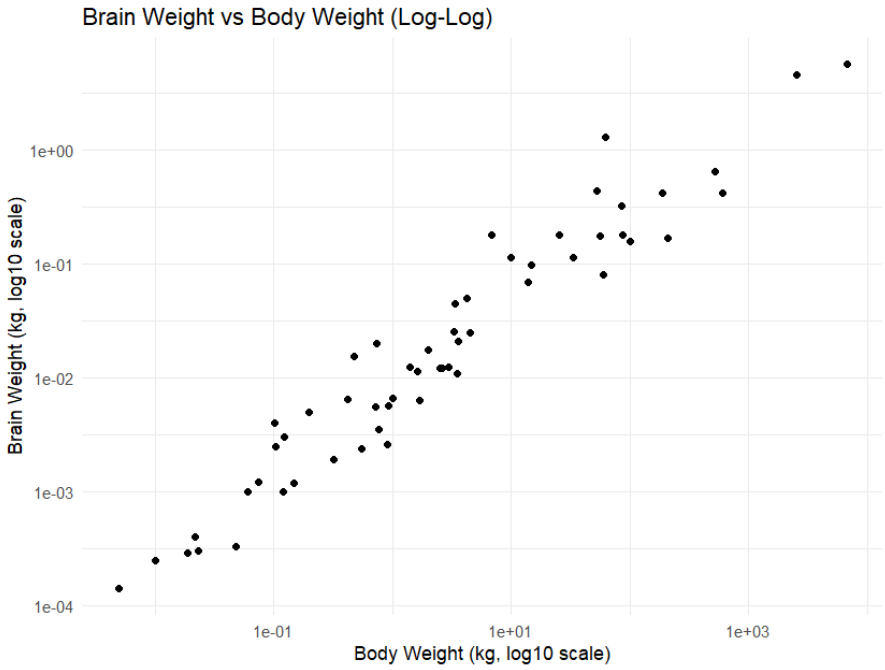
A coxcomb chart displays bar heights radially. It is visually appealing for categorical counts, but the area and angle can make exact comparisons harder than in a conventional bar chart.

Q3. Nonlinear Axes

(a) msleep brain weight vs body weight: linear and log-log

The linear plot shows raw body and brain weights. The log-log plot compresses large values and expands smaller values, making multiplicative relationships and scaling patterns easier to inspect. Log scales require positive values, so missing and nonpositive observations are removed.

```
library(ggplot2)
library(dplyr)
msleep2 <- msleep %>%
  filter(!is.na(body), !is.na(brain),
         body > 0, brain > 0)
# Linear axes
ggplot(msleep2, aes(x = body, y = brain)) +
  geom_point() +
  labs(title = "Brain Weight vs Body Weight (Linear)",
       x = "Body Weight (kg)", y = "Brain Weight (g)") +
  theme_minimal()
# Log-log axes
ggplot(msleep2, aes(x = body, y = brain)) +
  geom_point() +
  scale_x_log10() +
  scale_y_log10() +
  labs(title = "Brain Weight vs Body Weight (Log-Log)",
       x = "Body Weight (kg, log10 scale)",
       y = "Brain Weight (g, log10 scale)") +
  theme_minimal()
```

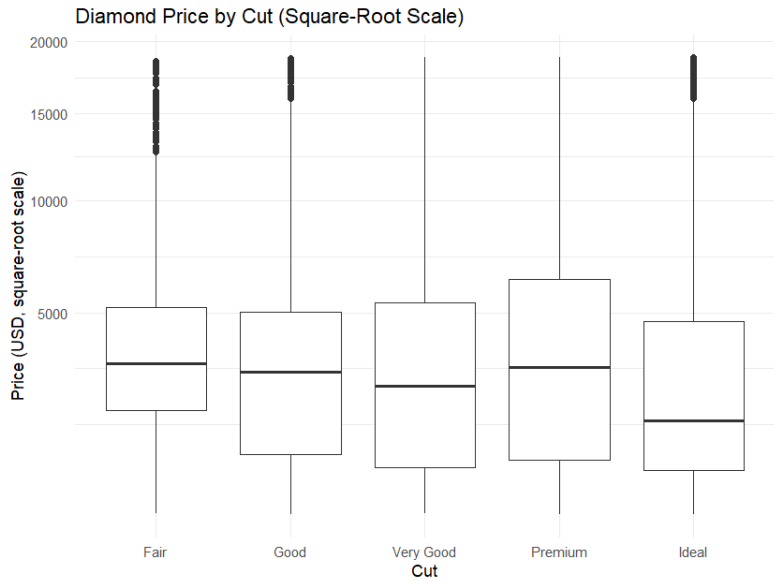


Comment: The log-log representation is generally more informative when body and brain weights span several orders of magnitude. It reduces skew and can make a power-law relationship appear approximately linear. The linear plot remains useful for interpreting absolute differences.

(b) diamonds price by cut: linear and square-root axes

```
library(ggplot2)
# Linear price axis
ggplot(diamonds, aes(x = cut, y = price)) +
  geom_boxplot() +
  labs(title = "Diamond Price by Cut (Linear Scale)",
       x = "Cut", y = "Price (USD)") +
  theme_minimal()

# Square-root price axis
ggplot(diamonds, aes(x = cut, y = price)) +
  geom_boxplot() +
  scale_y_sqrt() +
  labs(title = "Diamond Price by Cut (Square-Root Scale)",
       x = "Cut", y = "Price (USD, square-root scale)") +
  theme_minimal()
```

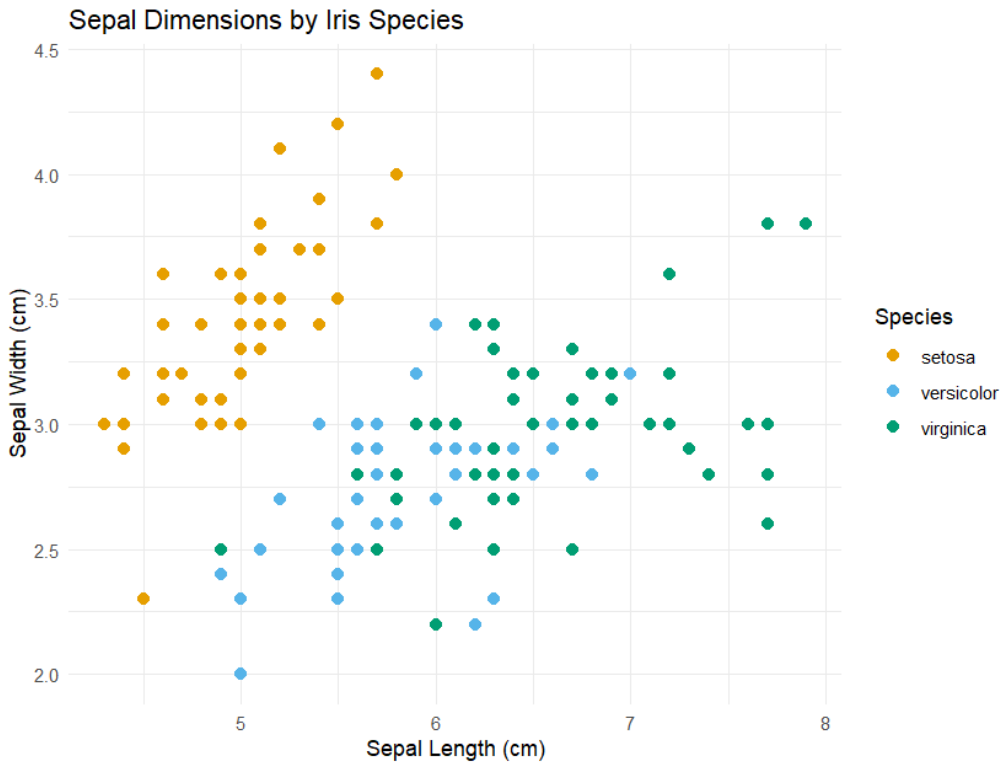


Comment: Diamond prices are right-skewed, with some very expensive observations. A square-root scale compresses high values less aggressively than a logarithmic scale and is useful for nonnegative data with moderate skew or when zero values may occur. It can make lower and middle prices easier to compare while retaining zero.

Q4. Color Scales: Qualitative, Sequential and Diverging

(a) Iris species with Okabe-Ito qualitative palette

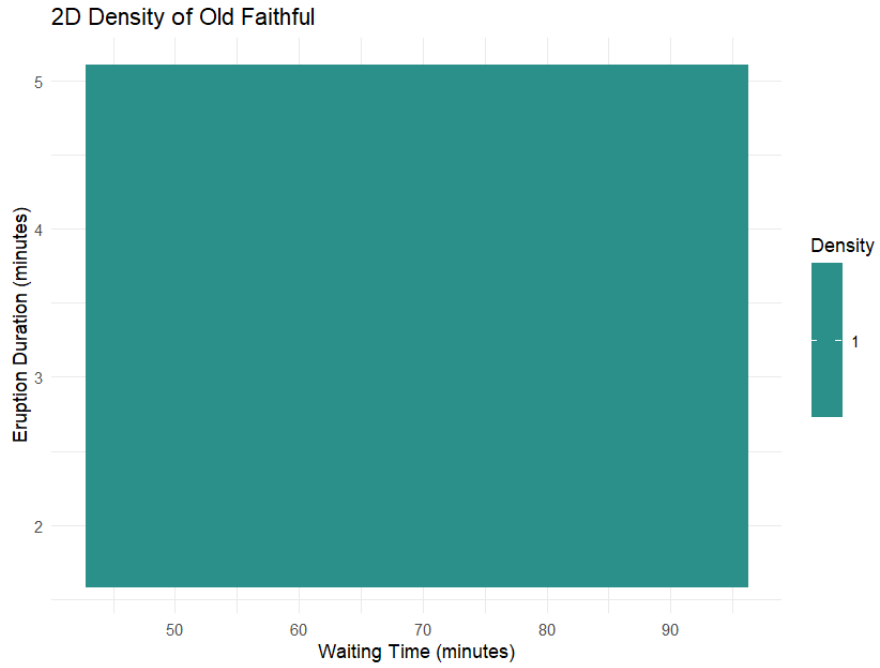
```
library(ggplot2)
okabe_ito <- c("#E69F00", "#56B4E9", "#009E73")
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width,
                 color = Species)) +
  geom_point(size = 2.5) +
  scale_color_manual(values = okabe_ito) +
  labs(title = "Sepal Dimensions by Iris Species",
       x = "Sepal Length (cm)", y = "Sepal Width (cm)",
       color = "Species") +
  theme_minimal()
```



The Okabe-Ito palette uses distinct, colorblind-safe hues to identify categories. It is a qualitative scale because colors distinguish species rather than encode magnitude.

(b) Faithful dataset 2D density heatmap with viridis

```
library(ggplot2)
ggplot(faithful, aes(x = waiting, y = eruptions)) +
  stat_density_2d(aes(fill = after_stat(level)),
                  geom = "raster", contour = FALSE) +
  scale_fill_viridis_c(name = "Density") +
  labs(title = "2D Density of Old Faithful",
       x = "Waiting Time (minutes)",
       y = "Eruption Duration (minutes)") +
  theme_minimal()
```



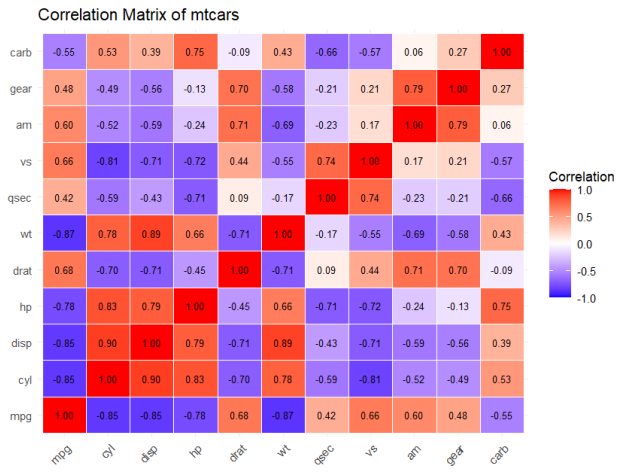
A sequential viridis scale represents increasing density with a perceptually ordered color progression. Viridis is designed to remain interpretable for many viewers, including people with common color-vision deficiencies, and is suitable for continuous nonnegative values.

(c) Correlation matrix of mtcars with diverging scale

```
library(ggplot2)
library(reshape2)

cor_mat <- cor(mtcars)
cor_long <- melt(cor_mat)

ggplot(cor_long, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = sprintf("%.2f", value)), size = 2.8) +
  scale_fill_gradient2(low = "blue", mid = "white",
                       high = "red", midpoint = 0,
                       limits = c(-1, 1), name = "Correlation") +
  labs(title = "Correlation Matrix of mtcars",
       x = NULL, y = NULL) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



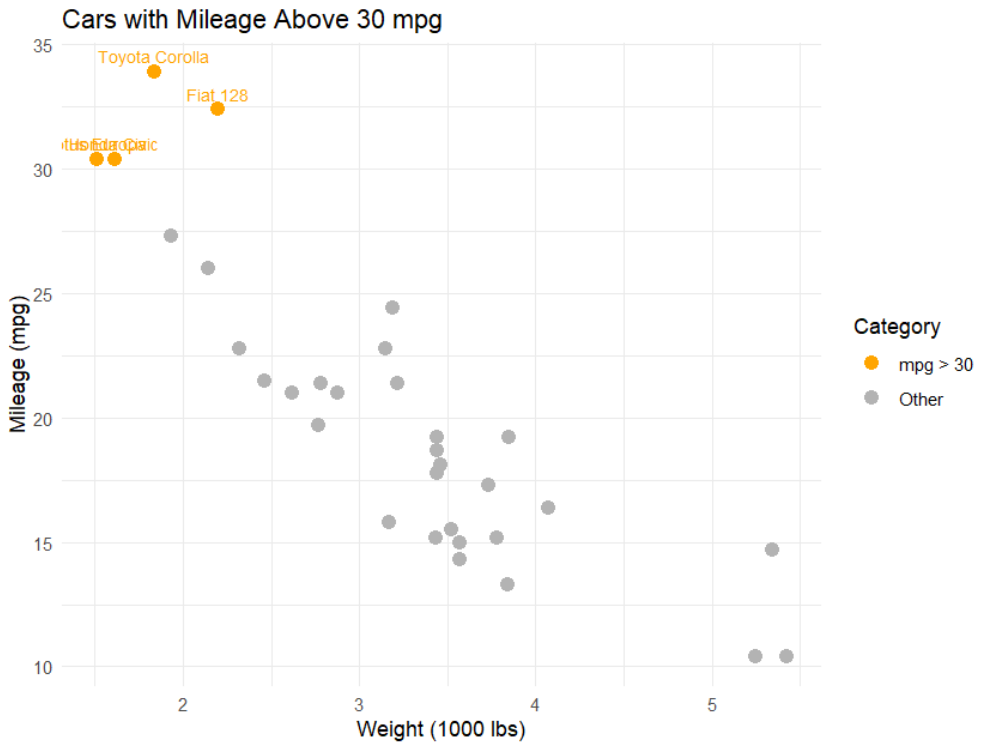
The diverging scale is centered at zero: negative correlations use one side of the palette, positive correlations use the other, and values near zero are neutral. Tile labels show the numerical correlation.

Q5. Color to Highlight and Continuous vs Discrete Scales

(a) Highlight cars with mpg > 30 and label them

Cars with mpg greater than 30 are shown in a bright accent color; all remaining cars are grey. The highlighted observations are labeled by row name (model name).

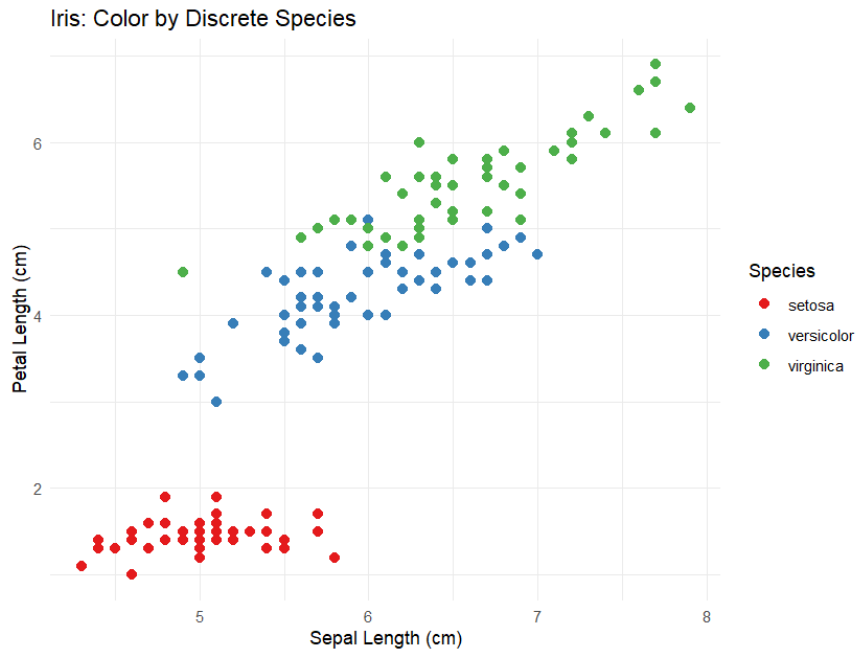
```
library(ggplot2)
mtcars2 <- mtcars
mtcars2$car <- rownames(mtcars2)
mtcars2$group <- ifelse(mtcars2$mpg > 30,
                        "mpg > 30", "Other")
ggplot(mtcars2, aes(x = wt, y = mpg, color = group)) +
  geom_point(size = 3) +
  geom_text(data = subset(mtcars2, mpg > 30),
            aes(label = car), vjust = -0.7,
            hjust = 0.5, size = 3, show.legend = FALSE) +
  scale_color_manual(values = c("mpg > 30" = "orange",
                                "Other" = "grey70")) +
  labs(title = "Cars with Mileage Above 30 mpg",
       x = "Weight (1000 lbs)", y = "Mileage (mpg)",
       color = "Category") +
  theme_minimal()
```



This highlighting technique directs attention to the subset that meets the condition while retaining the rest of the observations as context.

(b) Continuous Petal.Width color vs discrete Species color

```
library(ggplot2)
# Continuous color scale: numeric Petal.Width
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length,
                 color = Petal.Width)) +
  geom_point(size = 2.5) +
  scale_color_viridis_c(name = "Petal Width (cm)") +
  labs(title = "Iris: Color by Continuous Petal Width",
       x = "Sepal Length (cm)", y = "Petal Length (cm)") +
  theme_minimal()
# Discrete color scale: Species categories
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length,
                 color = Species)) +
  geom_point(size = 2.5) +
  scale_color_brewer(palette = "Set1", name = "Species") +
  labs(title = "Iris: Color by Discrete Species",
       x = "Sepal Length (cm)", y = "Petal Length (cm)") +
  theme_minimal()
```



Difference: A continuous color scale maps a numeric variable to a gradient, so intermediate colors represent intermediate Petal.Width values. A discrete color scale assigns separate colors to categories such as setosa, versicolor, and virginica; the colors identify group membership and do not imply numeric ordering.